

1) Find the interval of convergence of the foll. power series

a) $\frac{x}{1 \cdot 2} + \frac{x^2}{3 \cdot 4} + \frac{x^3}{5 \cdot 6} + \dots$

b) $x - \frac{x^2}{4} + \frac{x^3}{9} - \frac{x^4}{16} + \dots$

c) $x - \frac{x^2}{3} + \frac{x^3}{5} - \frac{x^4}{7} + \dots$

d) $\frac{x}{3 \cdot 1^2} + \frac{x^2}{3^2 \cdot 2^2} + \frac{x^3}{3^3 \cdot 3^2} + \frac{x^4}{3^4 \cdot 4^2} + \dots$

2) Check whether the foll. series is absolutely or conditionally convergent.

i) $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2n-1}$

ii) $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{\sqrt{n}}$

iii) $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n(n+1)}$

iv) $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2+1}$

Solutions:

i) a) $U_n = \frac{x^n}{(2n-1)(2n)}$ $\therefore U_{n+1} = \frac{x^{n+1}}{(2n+2-1)(2n+2)} = \frac{x^{n+1}}{(2n+1)(2n+2)}$

$$\left| \frac{U_{n+1}}{U_n} \right| = \left| \frac{x^{n+1}}{(2n+1)(2n+2)} \times \frac{(2n-1)(2n)}{x^n} \right| = |x| \times \frac{(2n-1)(2n)}{(2n+1)(2n+2)}$$

$$= |x| \times \frac{n^2 \left(2 - \frac{1}{n}\right) \times 2}{n^2 \left(2 + \frac{1}{n}\right) \left(2 + \frac{2}{n}\right)} = |x| \times \frac{\left(2 - \frac{1}{n}\right)}{\left(2 + \frac{1}{n}\right) \left(1 + \frac{1}{n}\right)}$$

$\therefore \lim_{n \rightarrow \infty} \left| \frac{U_{n+1}}{U_n} \right| = |x| \times \frac{2}{2 \times 1} = |x|$

$\therefore$ By Ratio Test, power series converges for $|x| < 1$
ie $-1 < x < 1$
 $\therefore$ Radius = 1.

i) For $x = -1$, series is $\sum \frac{(-1)^n}{(2n-1)(2n)}$ which is an alternating series with $U_n = \frac{1}{(2n-1)(2n)}$

$\therefore U_{n-1} = \frac{1}{(2n-2-1)(2n-2)} = \frac{1}{(2n-3)(2n-2)}$

$$U_n - U_{n-1} = \frac{1}{(2n-1)(2n)} - \frac{1}{(2n-3)(2n-2)}$$

$$= \frac{(2n-3)(2n-2) - 2n(2n-1)}{2n(2n-1)(2n-2)(2n-3)}$$

$$= \frac{4n^2 - 4n - 6n + 6 - 4n^2 + 2n}{2n(2n-1)(2n-2)(2n-3)} = \frac{-8n+6}{2n(2n-1)(2n-2)(2n-3)} < 0 \text{ for } n > 1$$

$\lim_{n \rightarrow \infty} U_n = \lim_{n \rightarrow \infty} \frac{1}{(2n-1)(2n)} = \lim_{n \rightarrow \infty} \frac{1}{2n^2 \left(2 - \frac{1}{n}\right)} = 0$

$\therefore$ By Leibnitz's test, series converges for $x = -1$.

ii) For $x=1$, the series is $\sum \frac{1}{(2n-1)(2n)}$.

$$u_n = \frac{1}{2n(2n-1)} = \frac{1}{n^2 \times 2 \left(2 - \frac{1}{n}\right)}$$

$$\text{let } v_n = \frac{1}{n^2}$$

$$\therefore \lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \lim_{n \rightarrow \infty} \frac{1}{2 \left(2 - \frac{1}{n}\right)} = \frac{1}{4} \neq 0$$

Also, $\sum v_n = \sum \frac{1}{n^2}$ is a p-series with $p=2 > 1$ $\therefore \sum v_n$ converges

$\therefore$ By comparison test, $\sum u_n$ also converges.

$\therefore$ series converges for $x=1$

$\therefore$ Interval of convergence is $-1 \leq x \leq 1$.

b) $x - \frac{x^2}{4} + \frac{x^3}{9} - \frac{x^4}{16} + \dots$

$$u_n = \frac{(-1)^{n-1} x^n}{n^2} \quad u_{n+1} = \frac{(-1)^n \cdot x^{n+1}}{(n+1)^2}$$

$$\left| \frac{u_{n+1}}{u_n} \right| = \left| \frac{(-1)^n x^{n+1}}{(n+1)^2} \times \frac{n^2}{(-1)^{n-1} x^n} \right| = \left| (-1) \cdot x \cdot \frac{n^2}{(n+1)^2} \right| = |x| \left(\frac{1}{1 + \frac{1}{n}} \right)^2$$

$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} |x| \left(\frac{1}{1 + \frac{1}{n}} \right)^2 = |x|$$

$\therefore$ By Ratio Test, power series converges for $|x| < 1$ i.e. $-1 < x < 1$.

(i) for $x=-1$, the series is $-1 - \frac{(-1)^2}{4} + \frac{(-1)^3}{9} - \frac{(-1)^4}{16} + \dots$

$$= -1 - \frac{1}{4} - \frac{1}{9} - \frac{1}{16} - \dots$$

$$= -\sum \frac{1}{n^2}$$

which is a p-series with $p=2 > 1$ $\therefore$ Series converges

ii) For $x=1$, series is $1 - \frac{1}{4} + \frac{1}{9} - \frac{1}{16} + \dots$

which is alternating with $u_n = \frac{1}{n^2}$.

$$\therefore u_{n-1} = \frac{1}{(n-1)^2}$$

$$\begin{aligned} u_n - u_{n-1} &= \frac{1}{n^2} - \frac{1}{(n-1)^2} = \frac{(n-1)^2 - n^2}{n^2(n-1)^2} = \frac{n^2 - 2n + 1 - n^2}{n^2(n-1)^2} \\ &= \frac{-2n+1}{n^2(n-1)^2} < 0 \text{ for } n > 1 \end{aligned}$$

$\lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \frac{1}{n^2} = 0$ $\therefore$ By Leibnitz's test, power series converges for $x=1$.

$\therefore$ Interval of convergence is $-1 \leq x \leq 1$

$$1) c) x - \frac{x^2}{3} + \frac{x^3}{5} - \frac{x^4}{7} + \dots$$

$$u_n = \frac{(-1)^{n-1} x^n}{2n-1} \quad u_{n+1} = \frac{(-1)^n x^{n+1}}{2(n+1)-1} = \frac{(-1)^n x^{n+1}}{2n+2-1} = \frac{(-1)^n x^{n+1}}{2n+1}$$

$$\left| \frac{u_{n+1}}{u_n} \right| = \left| \frac{(-1)^n x^{n+1}}{2n+1} \times \frac{2n-1}{(-1)^{n-1} x^n} \right| = \left| (-1) x \times \frac{2n-1}{2n+1} \right| = |x| \times \frac{n(2-\frac{1}{n})}{n(2+\frac{1}{n})}$$

$$\therefore \lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} |x| \frac{(2-\frac{1}{n})}{(2+\frac{1}{n})} = |x| \times \frac{2}{2} = |x|$$

$\therefore$ By Ratio test, power series converges for $|x| < 1$

(i) For $x = -1$, the series is $-1 - \frac{(-1)^2}{3} + \frac{(-1)^3}{5} - \frac{(-1)^4}{7} + \dots$ i.e., $-1 < x < 1$.

$$\begin{aligned} &= -1 - \frac{1}{3} - \frac{1}{5} - \frac{1}{7} - \dots \\ &= - \left(1 + \frac{1}{3} + \frac{1}{5} + \dots \right) \\ &= - \sum \frac{1}{2n-1} \end{aligned}$$

$$u_n = \frac{1}{2n-1} = \frac{1}{n(2-\frac{1}{n})}$$

$$v_n = \frac{1}{n}$$

$$\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \lim_{n \rightarrow \infty} \frac{1}{2-\frac{1}{n}} = \frac{1}{2} \neq 0$$

$\sum v_n = \sum \frac{1}{n}$ is a p-series with $p=1 \therefore \sum v_n$ diverges

$\therefore$ By comparison test, $\sum u_n = \sum \frac{1}{2n-1}$ also diverges

$\therefore - \sum \frac{1}{2n-1}$ also diverges.

$\therefore$ series diverges for $x = -1$.

(ii) for $x = 1$, the series is $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$ which is alternating with $u_n = \frac{1}{2n-1}$.

$$\therefore u_{n-1} = \frac{1}{2n-3}$$

$$u_n - u_{n-1} = \frac{1}{2n-1} - \frac{1}{2n-3} = \frac{2n-3 - (2n-1)}{(2n-1)(2n-3)} = \frac{-2}{(2n-1)(2n-3)}$$

$$\therefore u_n - u_{n-1} < 0 \text{ for } n > 1$$

$\lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \frac{1}{2n-1} = 0 \therefore$ By Leibnitz's test, series converges $\therefore$ Interval of convergence is $-1 < x \leq 1$.

$$d) \frac{x}{3 \cdot 1^2} + \frac{x^2}{3^2 \cdot 2^2} + \frac{x^3}{3^3 \cdot 3^2} + \frac{x^4}{3^4 \cdot 4^2} + \dots$$

$$\rightarrow u_n = \frac{x^n}{3^n n^2} \quad u_{n+1} = \frac{x^{n+1}}{3^{n+1} (n+1)^2} = \frac{x^n \cdot x}{3^n \cdot 3 (n+1)^2}$$

$$\left| \frac{u_{n+1}}{u_n} \right| = \left| \frac{x^n \cdot x}{3^n \cdot 3 (n+1)^2} \times \frac{3^n n^2}{x^n} \right| = \left| \frac{x}{3} \times \frac{n^2}{(n+1)^2} \right| = \frac{|x|}{3} \cdot \left(\frac{1}{1 + \frac{1}{n}} \right)^2$$

$$\therefore \lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \frac{|x|}{3}$$

$\therefore$ By Ratio Test, power series converges for $\frac{|x|}{3} < 1$

$$\text{i.e. } |x| < 3$$

$$\text{i.e. } -3 < x < 3$$

$\therefore$ Radius = 3

i) For $x = -3$, the series is $\sum \frac{(-3)^n}{3^n \cdot n^2} = \sum \frac{(-1)^n \cdot 3^n}{3^n \cdot n^2} = \sum \frac{(-1)^n}{n^2}$ which is alternating with $u_n = \frac{1}{n^2}$

$$u_n - u_{n-1} = \frac{1}{n^2} - \frac{1}{(n-1)^2} = \frac{(n-1)^2 - n^2}{n^2(n-1)^2} = \frac{n^2 - 2n + 1 - n^2}{n^2(n-1)^2} = \frac{-2n+1}{n^2(n-1)^2} < 0 \quad \text{for } n > 1$$

$$\lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \frac{1}{n^2} = 0$$

$\therefore$ By Leibnitz's test, series converges for $x = -3$.

ii) For $x = 3$, the series is $\sum \frac{3^n}{3^n n^2} = \sum \frac{1}{n^2}$ which is a p-series with $p = 2 > 1$ $\therefore$ series converges for $x = 3$

$\therefore$ Interval of convergence is $-3 \leq x \leq 3$.

$$2) \text{ (i) } \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2n-1}$$

Absolute series is

$$\sum |u_n| = \sum \left| \frac{(-1)^{n-1}}{2n-1} \right| = \sum \frac{1}{2n-1}$$

$$\therefore |u_n| = \frac{1}{2n-1} = \frac{1}{n(2 - \frac{1}{n})}$$

$$\text{let } v_n = \frac{1}{n}$$

$$\therefore \lim_{n \rightarrow \infty} \frac{|u_n|}{v_n} = \lim_{n \rightarrow \infty} \frac{1}{2 - \frac{1}{n}} = \frac{1}{2} \neq 0$$

$\therefore$

Also, $\sum v_n = \sum \frac{1}{n}$ is a p-series

with $p = 1$ $\therefore \sum v_n$ diverges.

$\therefore$ By comparison test $\sum |u_n|$ also diverges.

$\sum \frac{(-1)^{n-1}}{2n-1}$ is alternating series with

$$u_n = \frac{1}{2n-1}, \quad u_{n-1} = \frac{1}{2(n-1)-1} = \frac{1}{2n-3}$$

$$u_n - u_{n-1} = \frac{1}{2n-1} - \frac{1}{2n-3} = \frac{2n-3 - (2n-1)}{(2n-1)(2n-3)} = \frac{-2}{(2n-1)(2n-3)}$$

$$\therefore u_n - u_{n-1} < 0 \quad \text{for } n > 1$$

$$\lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \frac{1}{2n-1} = 0$$

$\therefore$ By Leibnitz's test, $\sum \frac{(-1)^{n-1}}{2n-1}$ converges

$\therefore \sum |u_n|$ diverges but $\sum u_n$ converges

$\therefore \sum \frac{(-1)^{n-1}}{2n-1}$ converges conditionally.

$$2 \text{ (ii)} \quad \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{\sqrt{n}}$$

$$\sum |u_n| = \sum \left| \frac{(-1)^{n-1}}{\sqrt{n}} \right| = \sum \frac{1}{\sqrt{n}} = \sum \frac{1}{n^{1/2}} \text{ is a } p\text{-series with}$$

$$p = \frac{1}{2} < 1$$

$\therefore \sum |u_n|$ diverges.

Now, $\sum \frac{(-1)^{n-1}}{\sqrt{n}}$ is an alternating series with $u_n = \frac{1}{\sqrt{n}}$.

$$\therefore u_{n-1} = \frac{1}{\sqrt{n-1}}$$

$$\text{Now, } n > n-1$$

$$\sqrt{n} > \sqrt{n-1}$$

$$\frac{1}{\sqrt{n}} < \frac{1}{\sqrt{n-1}}$$

$$\therefore u_n < u_{n-1}$$

$$\therefore u_n - u_{n-1} < 0 \quad \text{for } n > 1$$

$$\lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} = 0$$

$\therefore$ By Leibnitz's test $\sum \frac{(-1)^{n-1}}{\sqrt{n}}$ converges

$\therefore \sum u_n$ converges conditionally.

$$2 \text{ (iii)} \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n(n+1)}$$

$$\sum |u_n| = \sum \frac{1}{n(n+1)}$$

$$|u_n| = \frac{1}{n(n+1)} = \frac{1}{n^2(1+\frac{1}{n})}$$

$$v_n = \frac{1}{n^2}$$

$$\lim_{n \rightarrow \infty} \frac{|u_n|}{v_n} = \lim_{n \rightarrow \infty} \frac{1}{1+\frac{1}{n}} = 1 \neq 0$$

$\sum v_n = \sum \frac{1}{n^2}$ is a p-series with $p=2>1 \therefore \sum v_n$ converges

$\therefore$ By comparison test, $\sum |u_n|$ also converges.

$\therefore \sum \frac{(-1)^{n-1}}{n(n+1)}$ is absolutely convergent.

$$2 \text{ (iv)} \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2+1}$$

$$\sum |u_n| = \sum \frac{1}{n^2+1}$$

$$|u_n| = \frac{1}{n^2(1+\frac{1}{n^2})}$$

$$v_n = \frac{1}{n^2}$$

$$\lim_{n \rightarrow \infty} \frac{|u_n|}{v_n} = \lim_{n \rightarrow \infty} \frac{1}{1+\frac{1}{n^2}} = 1 \neq 0$$

$\sum v_n = \sum \frac{1}{n^2}$ which is a p-series with $p=2>1$

$\therefore \sum v_n$ converges.

$\therefore$ By comparison test, $\sum |u_n|$ also converges.

$\therefore \sum \frac{(-1)^{n-1}}{n^2+1}$ is absolutely convergent.